

Trajectory design strategies applied to temporary comet capture including Poincaré maps and invariant manifolds

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Abstract Temporary satellite capture (TSC) of Jupiter-family comets has been a focus of investigation within the astronomy community for decades. More recently, TSC has been approached from the perspective of dynamical systems theory, within the context of the circular restricted three-body problem (CR3BP). Thus, this problem serves as a testbed for exploring techniques that support trajectory design in similar dynamical regimes. In particular, an association between the invariant manifolds of libration point orbits and the paths of comets that experience TSC has been explored. In this investigation, TSC is further examined from the perspective of transit, that is, transition through the gateways associated with the collinear libration points, in the three-body problem. Periapsis Poincaré maps, previously employed for trajectory design in several investigations, are used to deliver insight into the nature of transit trajectories for energy levels near those associated with several Jupiter-family comets. The evolution of transit trajectories with increasing energy is explored, and the existence of solutions with similar characteristics to the paths of comets P/1996 R2, 82P/Gehrels 3, and 147P/Kushida–Muramatsu is demonstrated within the context of the planar CR3BP using planar periapsis maps. During TSC, the path of comet 111P/Helin–Roman–Crockett is highly inclined with respect to Jupiter; the motion of this comet is examined relative to invariant manifolds in the spatial CR3BP. A method to display the information contained in higher-dimensional Poincaré maps is also demonstrated, and is employed to locate a trajectory possessing the same qualitative characteristics as the path of 111P/Helin–Roman–Crockett.

Keywords Comets · Temporary satellite capture · Three-body problem · Poincaré map · Invariant manifolds · Periapsis map · Comet 111P/Helin–Roman–Crockett · Comet 147P/Kushida–Muramatsu · Comet 82P/Gehrels 3 · Comet P/1996 R2

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1 Introduction

Jupiter-family comets, due to their short orbital periods, frequently pass through perihelion, yielding a relatively large amount of observational data along their paths (Fernández 2005). Additionally, the orbital elements associated with these comets are distributed such that the likelihood of interaction between a member of the Jupiter-family of comets and the planet Jupiter is increased (Fernández 2005). Such interactions lead to temporary satellite capture (TSC) for several Jupiter-family comets, including comets 39P/Oterma, P/1996 R2, 82P/Gehrels 3, 147P/Kushida–Muramatsu, and 111P/Helin–Roman–Crockett. The phenomenon of TSC has been previously explored from the perspective of the circular restricted three-body problem (CR3BP) by a number of investigators. Heppenheimer (1975), Horedt (1976) and Heppenheimer and Porco (1977) examine the possibility of satellite capture through the L_1 gateway in the CR3BP. Koon et al. (2000) approach TSC from the perspective of dynamical systems theory, observing that certain features of the path of comet 39P/Oterma resemble characteristics of the invariant manifolds associated with planar Lyapunov orbits. These authors produce a heteroclinic–homoclinic chain that closely resembles the path of comet 39P/Oterma. Marchand (2000) and Howell et al. (2001b) examine the invariant manifold structures associated with periodic orbits in the spatial CR3BP; a combination of stable and unstable manifold arcs associated with halo orbits are located that closely resemble the trajectories of comets 39P/Oterma and 111P/Helin–Roman–Crockett. When examining comet behavior from the perspective of the invariant manifolds, a combination of both stable and unstable arcs are employed within a single path approximation due to the asymptotic nature of the manifolds. In this investigation, cometary motions are explored from the perspective of transit in the CR3BP using techniques previously demonstrated for trajectory design. Representing the paths of these comets as transit trajectories yields continuous arcs with no asymptotic features.

Trajectories that transition between adjoining regions within the zero-velocity curves (ZVCs) in the context of the CR3BP have been investigated by several researchers. Koon et al. (2000) and Gómez et al. (2004) examine transit in terms of the manifolds associated with libration point orbits in the vicinity of L_1 and L_2 . These manifolds act as separatrices, thus, transit solutions can be distinguished from non-transit trajectories by their location relative to the manifolds. In this investigation, transit and TSC are examined using periapsis Poincaré maps to display the structures associated with the invariant manifolds. The maps yield insight into the evolution of the solution space in the Sun–Jupiter system as the energy level is increased. Solutions possessing similar characteristics to the paths of comets P/1996 R2, 82P/Gehrels 3, and 147P/Kushida–Muramatsu are demonstrated using trajectory design techniques demonstrated by Koon et al. (2000) and employing planar periapsis maps. The task of locating three-dimensional trajectories that possess specific behaviors is non-trivial. Trajectory design techniques demonstrated by Gómez et al. (2004) are applied to examine the motion of comet 111P/Helin–Roman–Crockett relative to invariant manifolds in the spatial CR3BP. A strategy to represent the information contained in higher-dimensional periapsis maps computed in the spatial CR3BP is explored and is demonstrated for trajectory design in the spatial problem. This strategy is applied to locate a trajectory possessing the same qualitative characteristics as the path of 111P/Helin–Roman–Crockett.

2 Dynamical model

The evolution of short-period comet trajectories is perhaps most strongly influenced by the gravity of the Sun and Jupiter (Everhart and Nirupama 1970; Kresak 1979; Fernández

2005). Thus, the CR3BP is assumed as the dynamical model in this analysis, where the Sun and Jupiter are selected as the two primary bodies. While this represents a simplified model, previous investigations (e.g., Carusi and Valsecchi 1979) indicate that the influences of the Galilean satellites, as well as the outer planets, are relatively small. The motion of the comet (P_3), assumed to be a massless particle, is examined as it moves in the vicinity of the Sun (P_1) and Jupiter (P_2), each represented as a point mass. The orbits of the primary bodies are assumed circular relative to the system barycenter, B ; Heppenheimer (1975) and Horedt (1976) claim that the small eccentricity of Jupiter has a negligible effect on capture trajectories. A rotating frame, centered on B , is defined such that the rotating $\hat{x}$ -axis is directed from the Sun to Jupiter, the $\hat{z}$ -axis is parallel to the direction of the angular velocity of the primary system, and the $\hat{y}$ -axis completes the dextral orthonormal triad. The system of equations is non-dimensionalized using the following characteristic quantities: total mass, $m^* = m_1 + m_2 = 1.990420704855387 \times 10^{30}$ kg; the distance between the primaries, $\ell^* = R_1 + R_2 = 778412020$ km; and, characteristic time, $t^* = \sqrt{\frac{\ell^{*3}}{Gm^*}}$. Assuming cometary masses on the order of $m_3 \sim 10^{18}$ kg yields a non-dimensional mass $\frac{m_3}{m^*} \sim 5 \times 10^{-13}$ indicating that the assumption that the comet does not influence the motion of the primaries is reasonable. Defining the mass parameter $\mu = \frac{m_2}{m^*} = 9.538156685350 \times 10^{-4}$, the non-dimensional locations of the primaries defined in terms of rotating coordinates are $\mathbf{r}_1 = [-\mu, 0, 0]$, $\mathbf{r}_2 = [1 - \mu, 0, 0]$, where bold symbols represent vector quantities. The position vector of P_3 is written $\mathbf{r} = x\hat{x} + y\hat{y} + z\hat{z}$, and the first-order vector equation of motion is expressed as

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}), \quad (1)$$

where

$$\mathbf{f} = [\dot{x}, \dot{y}, \dot{z}, 2\dot{y} + U_x, -2\dot{x} + U_y, U_z]^T. \quad (2)$$

The pseudo-potential, U , is defined

$$U(x, y, z) = \frac{1 - \mu}{r_{13}} + \frac{\mu}{r_{23}} + \frac{1}{2}(x^2 + y^2), \quad (3)$$

where r_{i3} represents the distance between primary body P_i and the comet, P_3 . The quantities U_x, U_y, U_z represent partial derivatives of U with respect to rotating position coordinates. Five equilibrium points exist, and are depicted in Fig. 1a. The single, scalar integral of the motion, known as the Jacobi constant, C , is represented as $C = 2U(x, y, z) - v^2$, where $v = \sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2}$. From the expression for the Jacobi constant, the ZVCs are defined where $2U(x, y, z) - C = 0$ in the x - y plane. A sample schematic of a ZVC for a value of Jacobi constant such that $C_{L_3} < C < C_{L_2}$ appears in Fig. 1b. Here, the region surrounding the larger primary is labeled the *interior* region (I), the region in the vicinity of the smaller primary is defined as the P_2 region, and the *exterior* region (E) lies beyond the ZVCs.

3 Computation of stable and unstable invariant manifolds

Within the context of the planar CR3BP (PCR3BP), the invariant manifold “tubes” associated with the planar Lyapunov orbits act as separatrices and provide the mechanism for transport between adjoining regions of the ZVCs (Conley 1968; Koon et al. 2000; Wiggins et al. 2001; Gómez et al. 2004). In the spatial problem, both periodic and quasi-periodic orbits

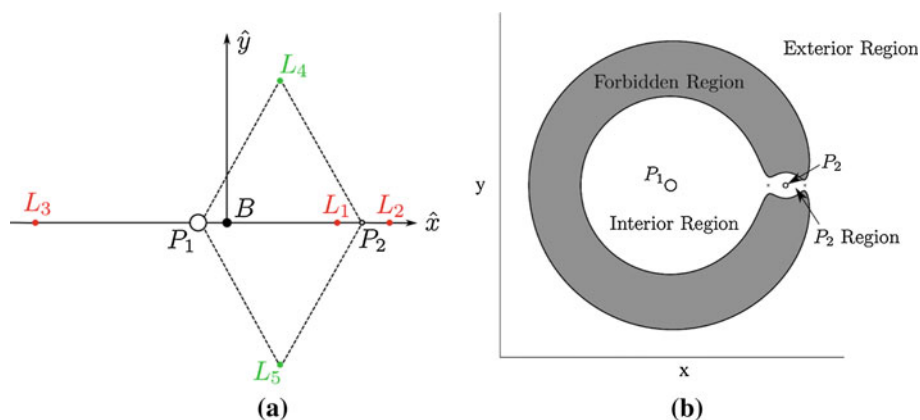
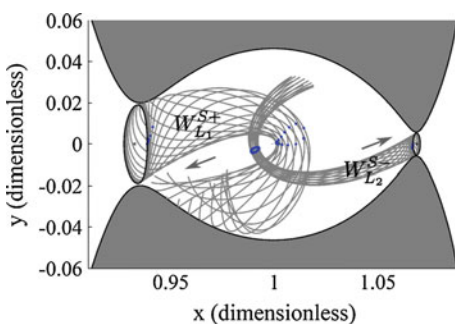


Fig. 1 Equilibrium points and ZVCs in the CR3BP, **a** schematic of libration points, **b** schematic of a sample ZVC

exist within the center manifold associated with the collinear libration points. Gómez et al. (2004) have demonstrated that the hyperbolic invariant manifold tangent to this center manifold provides the separatrix that distinguishes transit from nontransit behavior in the spatial problem. The periodic Lyapunov, vertical and halo orbits that exist in the vicinity of the collinear points may be computed numerically by exploiting their symmetries, as described by Grebow (2006). To compute the global manifolds associated with unstable periodic orbits, an initial step off the orbit onto a manifold is generated and is numerically propagated in the full nonlinear equations of motion. This step off the orbit is defined as a perturbation of the six-dimensional state of a fixed point, $\mathbf{x}^*$, along a periodic orbit (period T) in the direction of the stable/unstable eigenvector associated with $\mathbf{x}^*$. Assume that $\lambda^U > 1$ and $\lambda^S = 1/\lambda^U$ are the unstable and stable eigenvalues of the monodromy matrix, $\Phi(t_0 + T, t_0)$, corresponding to an unstable periodic orbit. Let $\mathbf{w}^U$ and $\mathbf{w}^S$ be their associated eigenvectors, and define $\mathbf{w}^{U+}$, $\mathbf{w}^{U-}$ ($\mathbf{w}^{S+}$, $\mathbf{w}^{S-}$) as the two directions associated with each eigenvector. The local half-manifold, W_{loc}^{U-} (W_{loc}^{S-}), is approximated by introducing a perturbation relative to $\mathbf{x}^*$ along the periodic orbit in the direction $\mathbf{w}^{U-}$ ($\mathbf{w}^{S-}$). Likewise, a perturbation relative to $\mathbf{x}^*$ in the direction $\mathbf{w}^{U+}$ ($\mathbf{w}^{S+}$) produces the local half-manifold W_{loc}^{U+} (W_{loc}^{S+}). The magnitude of the step along the direction of the eigenvector is denoted d , and the initial states along the local stable manifolds are computed as $\mathbf{x}^{S+} = \mathbf{x}^* + d \cdot \mathbf{w}^{S+}$, $\mathbf{x}^{S-} = \mathbf{x}^* - d \cdot \mathbf{w}^{S-}$, where $\mathbf{w}^{S+}$ and $\mathbf{w}^{S-}$ are normalized so that the vector containing the position components of the eigenvector is of unit length; this normalization provides a physical meaning for the value of d as a distance. The value of d is critical because it determines the accuracy with which the local manifolds are approximated. Selecting d too small yields manifold trajectories that require long integration times before departure from the vicinity of the periodic orbit, leading to accumulation of numerical error. If d is too large, then the approximation to the local manifold is poor. In this investigation, the value of d is selected so that propagating the initial state along the manifold back toward the periodic orbit, i.e., propagating $\mathbf{x}^{S-}/\mathbf{x}^{S+}$ in forward-time and $\mathbf{x}^{U-}/\mathbf{x}^{U+}$ in reverse-time, yields a manifold trajectory that completes at least two revolutions of the periodic orbit. For the Sun–Jupiter system, values of $d \sim 500$ km satisfy this requirement. The local stable manifolds are globalized by propagating the states $\mathbf{x}^{S+}$ and $\mathbf{x}^{S-}$ in reverse-time in the nonlinear model. This process yields the numerical approximation for the global manifolds W^{S+} and W^{S-} , respectively. The same procedure

Fig. 2 L_1 Lyapunov manifold, $W_{L_1}^{S+}$, and L_2 Lyapunov manifold, $W_{L_2}^{S-}$, appear in gray in the Sun–Jupiter system for $C = 3.0374$ with periaapses plotted in blue; Jupiter $2\times$ actual size



is employed to approximate the unstable global manifolds, W^{U+} and W^{U-} . A subset of the stable manifolds associated with planar Lyapunov orbits in the Sun–Jupiter system are plotted in gray in Fig. 2 with passages of perijove plotted in blue.

Several methods have been developed to locate the stable/unstable manifolds asymptotic to quasi-periodic orbits. To approximate the manifolds associated with quasi-periodic orbits in an ephemeris model, Barden (2000) and Howell et al. (2011a) assume the orbit is nearly periodic over two revolutions, and treat the state transition matrix over these two revolutions as a monodromy matrix. Computation of the stable/unstable manifolds then proceeds using the methods applied to periodic orbits. Marchand (2000) compares this strategy with the power method, employed by Gómez et al. (1993), in which the orbit is assumed to be periodic after some number, N , of revolutions. The state transition matrix associated with the orbit over N revolutions is employed as an approximation to the monodromy matrix, and the stable/unstable eigenvector directions tangent to the stable/unstable manifolds are computed. Gómez et al. (2005) compute quasi-periodic orbits in the vicinity of the collinear points, as well as the stable and unstable manifolds asymptotic to these orbits, by obtaining higher-order normal form expansions of the Hamiltonian to decouple the center and hyperbolic manifolds associated with the libration point of interest. In this study, families of quasi-periodic tori and their associated manifolds are computed numerically using techniques demonstrated by Olikara and Scheeres (2012). Assuming the function $\mathbf{v}(\theta_0, \theta_1)$ describes a two-dimensional torus on which a quasi-periodic orbit, with associated frequencies $\omega_0 = \dot{\theta}_0$, $\omega_1 = \dot{\theta}_1$, lies, then, the dimension of the torus may be reduced to one by selecting an initial value of θ_0 so that an invariant circle, $\mathbf{u}(\theta_1)$, along the torus is defined. A map, $\mathbf{G}$, is defined based on the frequencies ω_0, ω_1 so that propagating discretized states along $\mathbf{u}(\theta_1)$ for time $T_0 = \frac{2\pi}{\omega_0}$ and removing the rotation by angle $\rho = \omega_1 \cdot T_0$ due to frequency ω_1 yields $\mathbf{G}(\mathbf{u}) = \mathbf{u}$. To compute a torus, a differential corrections algorithm is employed to locate ω_0, ω_1 , and discretized states along $\mathbf{u}(\theta_1)$ that satisfy $\mathbf{G}(\mathbf{u}) - \mathbf{u} = 0$, while applying an additional constraint on the value of Jacobi constant. Once a torus is located, pseudo-arclength continuation provides a method to locate additional tori in a family, assuming that additional phase constraints on θ_0 and θ_1 are included. The families of tori corresponding to quasi-halo and Lissajous orbits are initialized from the periodic halo and vertical orbits by employing the associated stability information to locate a linear approximation to a nearby invariant circle. Recalling that a periodic orbit represents a fixed point under the stroboscopic map $\mathbf{F}(\mathbf{x})$ defined by time T , then stability information for the periodic orbit is recovered by examining the eigenvalues of the linearization of the map, i.e., the monodromy matrix $\Phi(T, 0) = \mathbf{F}_x$. Analogously, the invariant circle, $\mathbf{u}(\theta_1)$, represents a fixed point of the map $\mathbf{G}(\mathbf{u})$ defined by time T_0 and frequency ω_1 . Thus, stability of the torus is determined by the eigenvalues of the matrix $\mathbf{G}_x$; the eigenvectors corresponding to eigenvalues that lie off of

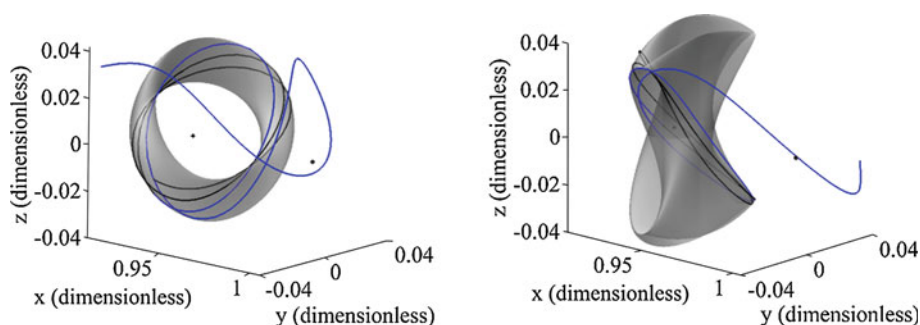


Fig. 3 Quasi-periodic tori with sample stable manifolds (blue) in the Sun–Jupiter system for $C = 3.029$; Jupiter $10\times$ actual size

the unit circle in the complex plane provide the directions tangent to the stable and unstable manifolds associated with each of the discretized states along $\mathbf{u}(\theta_1)$ on the torus. Details on the computation of quasi-periodic tori and their associated manifolds are provided by [Olikara and Scheeres \(2012\)](#). Examples of tori corresponding to quasi-halo (left) and Lissajous (right) orbits appear in Fig. 3. For each torus, a single stable manifold is displayed in blue; propagating the manifolds in forward-time (back toward the quasi-periodic orbits) for $2 \cdot T_0$ yields two revolutions along the quasi-periodic orbits, plotted in black. Alternatively, numerical methods exist to compute quasi-periodic orbits over a finite time interval. Assuming that the orbit is nearly periodic over a particular revolution, i.e., that the initial and final states are sufficiently close, the period, T , necessary to compute the monodromy matrix for periodic orbits, may be approximated. For example, T may be taken as the time elapsed between the initial state along a particular revolution and the closest approach to that state within one revolution of the quasi-periodic orbit. With an approximation for the monodromy matrix computed (for one revolution of the quasi-periodic orbit), approximations to the global stable and unstable manifolds may be located using the same methodology as for periodic orbits. Employing this method to compute the manifolds associated with quasi-periodic orbits yields qualitatively similar results when compared with the method defined by Olikara and Scheeres for all results in this study.

4 Periapsis surface of section

The use of a Poincaré section, in addition to a constraint on the value of Jacobi constant, reduces the dimension of the system by two. In the PCR3BP, the state space is, therefore, entirely represented by the projection onto a plane. The periapsis surface of section, demonstrated by [Villac and Scheeres \(2003\)](#) and [Paskowitz and Scheeres \(2006\)](#), can be projected and viewed in configuration space rather than a mixed position-velocity space, allowing for an intuitive visualization of the map. The periapsis map also reveals distinct regions associated with immediate escape from the smaller primary. In the spatial CR3BP (SCR3BP), the reductions obtained from constraining Jacobi constant and employing a Poincaré map yield a four-dimensional system. Thus, the state space is no longer fully represented by a planar projection, however, the periapsis surface of section still proves to be a useful tool for the visualization of the dynamics in the SCR3BP.

The periapsis surface of section is defined such that the radial velocity, $\dot{r}$, of P_3 relative to the primary is zero, and radial acceleration is positive, i.e., $\ddot{r} > 0$. As an example, the

periapsis structures associated with invariant manifold tubes corresponding to *planar* L_1 and L_2 Lyapunov orbits for a specified Jacobi constant value appear in Fig. 2. The manifolds are propagated through a finite time interval, and all passages of perijove that occur within the region $x_{L_1} \leq x \leq x_{L_2}$ are indicated as blue points. Some, but not all, of the manifolds intersect the $\dot{r} = 0$ hyperplane near the gateways. These manifolds undergo two passages of perijove (one near the gateway and one near Jupiter) during the specified time interval while the others experience only one periapsis near Jupiter. By previous convention (Koon et al. 2000; Gómez et al. 2004), let $\Gamma_{L_i,m}^S$ denote the contour formed by the periapses along the m th intersection of the L_i Lyapunov orbit stable manifold tube with the periapsis surface in the P_2 region, and $\Gamma_{L_j,n}^U$ denote the periapsis contour formed by the n th intersection of the L_j Lyapunov orbit unstable manifold tube. Then, the blue points in Fig. 2 form closed contours $\Gamma_{L_1,1}^S$ and $\Gamma_{L_2,1}^S$, neglecting the periapses that occur in the vicinity of the Lyapunov orbits. If all perijoves within the range $x_{L_1} \leq x \leq x_{L_2}$ are considered, including those near the Lyapunov orbits, then several of the blue points in Fig. 2 actually belong to contours $\Gamma_{L_1,2}^S$ and $\Gamma_{L_2,2}^S$, and the contours are no longer closed.

A map corresponding to all periapses along $W_{L_1}^{S+}$ within a roughly 500-year simulation (265 nondimensional time units; 1 year indicates 1 revolution of the Earth about the Sun) appears in Fig. 4a; periapses that define the manifold contours $\Gamma_{L_1,1 \sim N}^S$ are plotted in blue ($N = 175$ for this time interval). To examine the evolution of a large number of initial states at this Jacobi level, a batch of escape trajectories is obtained by propagating a grid of about 30,000 apse points within the P_2 region for 500-year in forward-time. Those trajectories that cross the boundaries $x = x_{L_2} - 0.05$ and $x = x_{L_2} + 0.05$, where $0.05 \sim 3.9 \times 10^7$ km, are defined as L_1 and L_2 escape trajectories, respectively. In Fig. 4, the periapses along escape trajectories are colored to reflect the number of periapses prior to escape. For example, periapses colored orange reflect the final perijove prior to escape through the L_1 gateway. In total, the colored clusters of points in Fig. 4a represent locations of the last six periapses before escape along the L_1 forward-time escape trajectories, and are denoted $\Pi_{L_1,1-6}^{t+}$, where superscript $t+$ indicates forward-time escape. The map representing periapses along L_1 reverse-time escape trajectories (temporary captures) and the unstable manifolds is simply the reflection of the map from Fig. 4a across the $\hat{x}$ -axis. The map in Fig. 4b depicts all periapses along $W_{L_2}^{S-}$ within the 500-year reverse-time propagation ($\Gamma_{L_2,1 \sim N}^S$, where $N = 174$ for the given interval), as well as the last six periapses before escape along L_2 forward-time escape trajectories, denoted $\Pi_{L_2,1-6}^{t+}$. The green, yellow, and gray periapses, i.e., $\Pi_{L_2,4-6}^{t+}$, fill very narrow regions that are essentially overlapped by the manifold periapses on the map. Reflecting this map across the $\hat{x}$ -axis yields the map depicting periapses along the manifolds $W_{L_2}^{U-}$ and L_2 reverse-time escape trajectories. The delineation between regions of allowed periapsis and apoapsis occurs where $\dot{r} = 0$, and is plotted as dashed black lines in Fig. 4.

Due to the complicated behavior of the manifold periapses as well as the fact that the transit periapsis colors correspond to the number of perijoves prior to escape or after capture, the periapsis count requires an additional note. Recall that only a subset of the manifolds in Fig. 2 encounter their first perijove near the gateways; thus, only a portion of the manifold tube crosses the $\dot{r} = 0$ hyperplane. The manifold tubes bound the regions of transit, thus, only some of the transits experience the additional passage of perijove. When counting periapses prior to escape along the transit trajectories, the subset of transits that do experience this additional periapsis possess a shifted counting number and the transit periapses within a particular region may not be colored uniformly. The map in Fig. 5a represents a zoomed view of the map in Fig. 4a; some cyan periapses occur within the orange region, some magenta periapses within the cyan region, etc. Clearly, when coloring transit periapses to correspond to the number of

Fig. 4 Planar manifold periapsis structures in the Sun–Jupiter system for $C = 3.0374$; Jupiter $10\times$ actual size, **a** first six periapses along $W_{L_1}^{S+}$ and along L_1 forward-time escapes, **b** first six periapses along $W_{L_2}^{S-}$ and along L_2 forward-time escapes

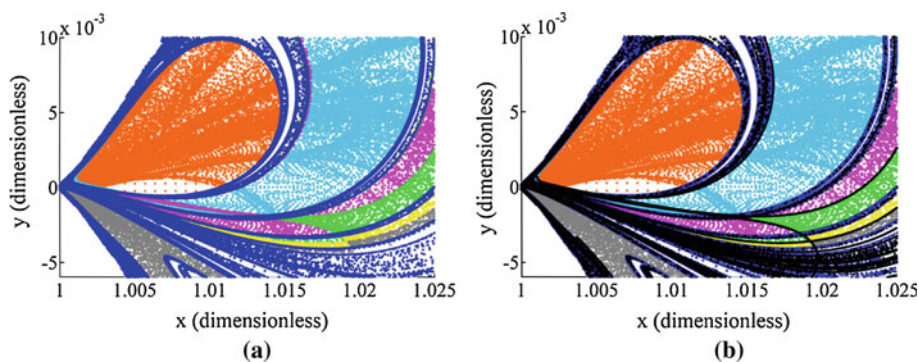
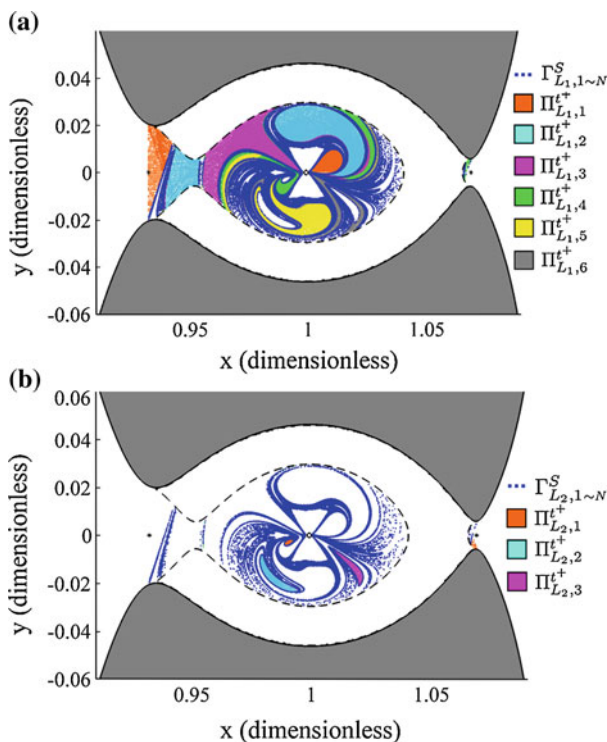


Fig. 5 Additional boundaries between colored regions $\Pi_{L_1,1-6}^{t+}$ are supplied by the periapses along trajectories that experience $\dot{r} = \ddot{r} = 0$, at some time, **a** zoomed view of the map in Fig. 4a, **b** periapses along trajectories that experience $\dot{r} = \ddot{r} = 0$ plotted in black

periapses prior to escape (or after capture), as in Fig. 4, an additional boundary is necessary to distinguish differently colored regions within the manifold periapsis contours. This additional boundary, plotted in black in Fig. 5b, is supplied by the periapses along trajectories that, at some time, yield an apse for which $\ddot{r} = 0$. The boundary that distinguishes differently colored regions within the manifold periapsis contours is obtained from the periapses along trajectories that, at some time, possess an apse for which $\ddot{r} = 0$. To produce the black periapses in Fig. 5b, a batch of initial conditions is generated along the $\dot{r} = \ddot{r} = 0$ contour.

Because the transit periapses are counted backward from the escape event, only the periapses that occur *prior* to the apse with $\dot{r} = 0$ will have shifted counting numbers. Thus, propagating the $\dot{r} = \ddot{r} = 0$ initial conditions in reverse-time yields the periapses that form the additional boundary necessary to distinguish regions $\Pi_{L_1,n}^{t+}$, plotted in black in Fig. 5b. (Note that the boundary distinguishing regions $\Pi_{L_1,n}^{t-}$ along reverse-time escapes is formed by propagating the $\dot{r} = \ddot{r} = 0$ initial conditions in forward-time and recording periapsis passages.)

5 Transit and temporary capture in the planar CR3BP

Transit trajectories, solutions that transition between adjoining regions within the ZVCs, have been examined by a number of investigators. Conley (1968), Koon et al. (2000), and Gómez et al. (2004) demonstrate that the invariant manifolds associated with libration point orbits in the CR3BP are separatrices that govern transit through the L_1 and L_2 gateways. Koon et al. (2000) demonstrate the use of invariant manifolds and Poincaré maps to locate planar transit trajectories as well as heteroclinic connections between planar Lyapunov orbits. These authors also explore the role of invariant manifolds in the resonance transition experienced by comet 39P/Oterma. In the SCR3BP, Gómez et al. (2004) compute the stable and unstable manifolds asymptotic to the center manifolds associated with the collinear points and employ Poincaré maps to generate transit trajectories in the spatial CR3BP. The use of invariant manifolds and Poincaré maps is also demonstrated by these authors to compute heteroclinic connections between libration point orbits in the spatial problem. Villac and Scheeres (2003) and Paskowitz and Scheeres (2006) use periapsis maps to classify trajectory behavior, including transit, in the Hill problem through numerical simulation of periapsis initial conditions in the vicinity of P_2 . Howell et al. (2012) demonstrate the use of periapsis maps for various mission design scenarios. In this investigation, the periapsis surface of section is incorporated and the resulting maps are viewed in configuration space.

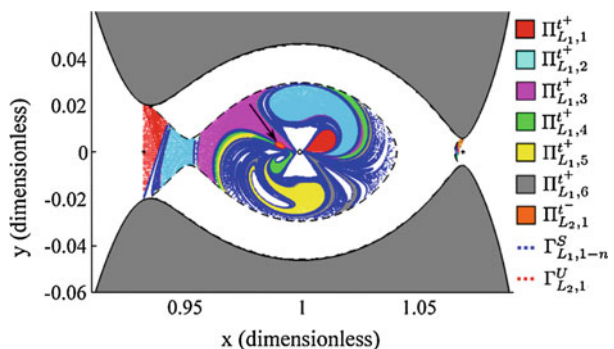
5.1 Identification of temporary capture trajectories possessing specified characteristics in the PCR3BP

Computation of transit solutions with predetermined characteristics, e.g., entering/exiting a specified gateway and possessing a predetermined number of revolutions about P_2 , has been previously demonstrated through the use of Poincaré maps (Koon et al. 2000; Gómez et al. 2004). For an initial state inside a contour $\Gamma_{L_i}^S(\Gamma_{L_i}^U)$, the trajectory enters the interior ($i = 1$) or exterior ($i = 2$) region from the P_2 region through the L_i gateway in forward-time (reverse-time). Recall that I and E represent the interior and exterior regions of the ZVCs. Using maps to display both $\Gamma_{L_i}^S$ and $\Gamma_{L_j}^U$, it is possible to design trajectories with the following predetermined behaviors: (1) long-term capture trajectories (initial state is outside $W_{L_1}^{U+}$, $W_{L_1}^{S+}$, $W_{L_2}^{S-}$, and $W_{L_2}^{S+}$) (Haapala 2010; Davis and Howell 2011; Haapala and Howell 2011; Davis and Howell 2012); (2) I-to-I transit (initial state is within $\Gamma_{L_1}^U \cap \Gamma_{L_1}^S$); (3) E-to-E transit (initial state is within $\Gamma_{L_2}^S \cap \Gamma_{L_2}^U$); and (4) I-to-E/E-to-I transit (initial state is within $\Gamma_{L_1}^U \cap \Gamma_{L_2}^S / \Gamma_{L_1}^S \cap \Gamma_{L_2}^U$). Through the use of Poincaré maps, computation of long-term capture trajectories and transit orbits is straightforward.

To compute a transit trajectory with predetermined characteristics, an initial state is selected within $\Gamma_{L_i,m}^S \cap \Gamma_{L_j,n}^U$ (equivalently, $\Pi_{L_i,m}^{t+} \cap \Pi_{L_j,n}^{t-}$). Propagating this state in both forward- and reverse-time, a transit orbit, γ , passing through the L_i gateway in forward-time and the L_j gateway in reverse-time is produced. Defining one revolution about Jupiter to be comprised of one perijove and one apojuve, γ experiences $m+n-1$ periapses and $m+n-2$ apoapses. Then, the number of revolutions, p , about Jupiter is evaluated as $p = m+n-3/2$. To design transit trajectories with specified behavior in the vicinity of Jupiter, the contours $\Gamma_{L_i,m}^S$ and $\Gamma_{L_j,n}^U$ and the regions $\Pi_{L_i,m}^{t+}$ and $\Pi_{L_j,n}^{t-}$ are selected such that $m+n = p+3/2$. As an example, to locate E-to-I transits in the Sun–Jupiter system at an energy level of $C = 3.0374$, the map appearing in Fig. 6 is employed. Stable manifold contours $\Gamma_{L_1,1 \sim N}^S$ (blue) and the contour depicting the first perijoves along the unstable manifold $\Gamma_{L_2,1}^U$ (red, indicated by arrow) appear, as well as regions $\Pi_{L_1,1-6}^{t+}$ and $\Pi_{L_2,1}^{t-}$. A zoomed view of contour $\Gamma_{L_2,1}^U$ appears in Fig. 7a. An intersection of contours $\Gamma_{L_2,1}^U$ and $\Gamma_{L_1,m}^S$ (regions $\Pi_{L_2,1}^{t-}$ and $\Pi_{L_1,m}^{t+}$) does not exist for $m \leq 2$, thus, for this system and value of Jacobi constant, E-to-I (and I-to-E) transits corresponding to $p = 0.5, 1.5$ do not exist. Regions $\Pi_{L_2,1}^{t-}$ (orange) and $\Pi_{L_1,3-5}^{t+}$ (magenta, green, yellow) do overlap, however, yielding transits of 2.5, 3.5, 4.5 revolutions. A representative $p = 2.5$ E-to-I transit, generated from the periapsis initial condition marked in magenta in Fig. 7a, is over-plotted on the map in Fig. 7b.

As the energy level is increased, the P_2 region expands and a larger percentage of the initial conditions lead to escape from the vicinity of Jupiter. With increasing energy, the structures in the periapsis maps change, as demonstrated in Fig. 8, where maps depicting L_1 escape and L_2 capture periapses appear for $C = 3.035$, as well as $C = 3.025$ (a value close to that associated with the comet P/1996 R2 during TSC). One result of varying C is a change in the range of available p values for transit trajectories. For example, the periapsis map depicting $\Pi_{L_1,1-2}^{t+}$ and $\Pi_{L_2,1}^{t-}$ for $C = 3.035$ appears in Fig. 9a. The regions $\Pi_{L_2,1}^{t-}$ and $\Pi_{L_1,2}^{t+}$ now overlap, allowing the existence of E-to-I transits of 1.5 revolutions; a sample trajectory is overplotted on the map. Further decreasing C to a value of 3.025 introduces an overlap of regions $\Pi_{L_2,1}^{t-}$ and $\Pi_{L_1,1}^{t+}$ and $p = 0.5$ E-to-I transits are available, as demonstrated in Fig. 9b. For the map of L_1 escapes appearing in Fig. 8a, b, regions $\Pi_{L_1,1-5}^{t+}$ and $\Pi_{L_2,1-5}^{t-}$ are visible, however, the regions $\Pi_{L_1,6}^{t+}$ and $\Pi_{L_2,6}^{t-}$ are so narrow that they are almost completely covered by Lyapunov manifold periapses. Regions $\Pi_{L_1,3-6}^{t+}$ and $\Pi_{L_2,3-6}^{t-}$ lie in thin regions, nearly covered by the manifold periapses, and are no longer visible in Fig. 8c, d. Clearly, as the energy level

Fig. 6 Periapsis map employed for locating E-to-I transit trajectories in Sun–Jupiter system for $C = 3.0374$; Delineation between regions of allowed periapsis and apoapsis appears as dashed black line; Jupiter $10 \times$ actual size



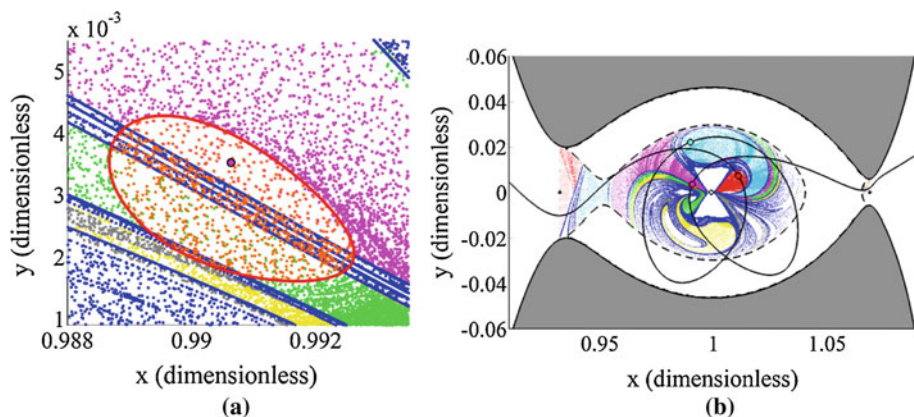


Fig. 7 Transit orbit about Jupiter corresponding to $p = 2.5$ appears with periaapses marked as dots; Jupiter $10\times$ actual size, **a** close view of $\Gamma_{L_{2,1}}^U$, **b** $p = 2.5$ E-to-I transit

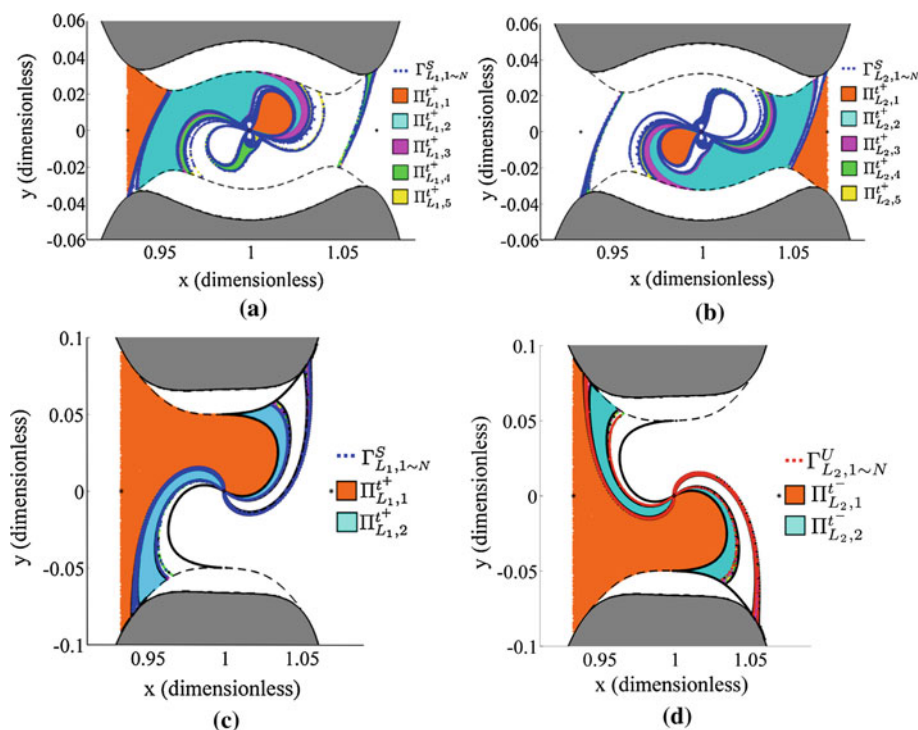


Fig. 8 Periaipsis regions associated with escape for varying values of Jacobi constant; Jupiter $10\times$ actual size, **a** $C = 3.035L_1$ escapes, **b** $C = 3.035L_2$ captures, **c** $C = 3.025L_1$ escapes, **d** $C = 3.025L_2$ captures

is increased and the gateways enlarge, the regions defined by the first crossings of transit trajectories expand. A greater number of trajectories escape before experiencing subsequent passages of perijove, thus, regions $\Pi_{L_{1,n}}^{t+}$ and $\Pi_{L_{2,n}}^{t-}$ decrease in size as n increases. Although some trajectories switch between direct and retrograde motion during the simulations, only prograde periaapses are plotted in these maps.

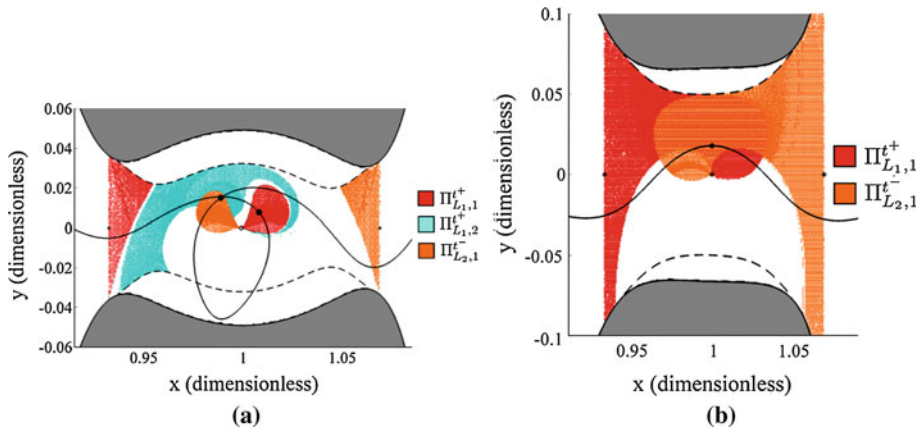


Fig. 9 Decreasing the Jacobi constant admits transit orbits characterized by $p < 2.5$ revolutions about Jupiter; Jupiter $10\times$ actual size, **a** $C = 3.035$, $p = 1.5$, **b** $C = 3.025$, $p = 0.5$

5.2 Analysis of comet paths from the perspective of the PCR3BP

Several Jupiter-family comets experience TSC, i.e., transit between the interior/exterior region and the Jupiter region. For comets that experience at least one passage of perijove, periapsis maps offer a tool for the analysis of cometary behaviors. Although the comet trajectories are not planar, investigation of comet perijoves relative to the planar periapsis maps yields preliminary insight into the cometary motion. Comets that have experienced TSC include P/1996 R2 (R2), 82P/Gehrels 3 (Gehrels), and 147P/Kushida–Muramatsu (KM); the ephemeris paths of these comets in the vicinity of Jupiter, obtained from the JPL HORIZONS database, are transformed into the Sun–Jupiter rotating frame and are plotted in black in Figs. 10a, b, and 11b with periapses plotted as red points. Each path switches between direct and retrograde motion during TSC, however, the heliocentric orbits remain prograde. The value of C fluctuates along the ephemeris paths; approximate values appear in the captions in Figs. 10a, b, and 11b. For the high energies associated with these comets, the gateways at L_1 and L_2 are widely opened, clearly facilitating escape from the vicinity of Jupiter. It is useful, then, to explore these trajectories from the perspective of Poincaré maps to determine why the comets do not escape immediately, but remain captured temporarily.

Planar periapsis maps associated with the energy levels representative of the comets are computed by identifying the type of transit (i.e., E-to-I, I-to-I, etc.) and the number of revolutions about Jupiter during TSC for each comet. Comets R2 and Gehrels are each E-to-I transits, experiencing 2.5 and 1.5 revolutions about Jupiter, considering only apses within the range $x_{L1} \leq x \leq x_{L2}$. The number of revolutions about Jupiter may not be obvious from Fig. 10a, b, but a useful definition computes p as equal to $\frac{1}{2}(\# \text{ periapses} + \# \text{ apoapses})$. To examine the behavior of these comets relative to the stable L_1 and unstable L_2 Lyapunov manifolds, maps depicting periapses along these manifolds for an energy level ($C = 3.025$) near those of the comets appear in Fig. 8c, d for a 500-year simulation; regions $\Pi_{L1,1-3}^{t+}$ and $\Pi_{L2,1-3}^{t-}$ are also displayed. Maps corresponding to $C = 3.0225$, a closer approximation for the energy level of R2 during TSC, appear very similar to those in Fig. 8c, d, where the black periapses in the maps correspond to periapses along trajectories that experience $\dot{r} = \ddot{r} = 0$ at some time, as discussed in Sect. 4 and demonstrated in Fig. 5b. Consider the last perijove before escape, i.e., the third perijove after entry, along the path of comet R2. In the context of

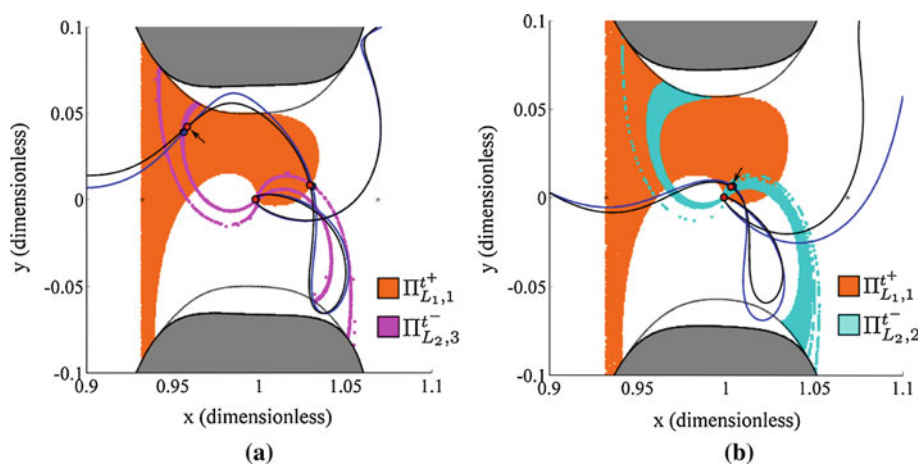


Fig. 10 Periapsis maps associated with energy levels near those of comets P/1996 R2 and 82P/Gehrels 3 with ephemeris paths in *black* with *red* periapses; sample PCR3B paths in *blue* with *blue* periapses, **a** path of P/1996 R2 ($C = 3.025$), **b** Path of 82P/Gehrels 3 ($C = 3.0225$)

the PCR3BP, it is expected that this perijove will lie within the intersection $\Pi_{L_{1,1}}^{t+} \cap \Pi_{L_{2,3}}^{t-}$; these regions are plotted in addition to the ephemeris path of R2 in Fig. 10a. The planar projection of the comet perijove location indicated with the black arrow nearly overlaps the intersection $\Pi_{L_{1,1}}^{t+} \cap \Pi_{L_{2,3}}^{t-}$. To produce a sample path in the PCR3BP that possesses similar behavior to that of comet R2, a periapsis within this intersection is selected and is propagated, yielding the blue trajectory with blue periapses in Fig. 10a (the red periapses along the ephemeris path overlap the blue periapses in some cases). A similar process is employed to examine the path of comet Gehrels. The only prograde perijove along the path is the last periapsis before escape, thus, the regions $\Pi_{L_{1,1}}^{t+}$, $\Pi_{L_{2,2}}^{t-}$ are plotted in Fig. 10b. Again, the planar projection of the indicated perijove location lies nearly within the intersection of the colored regions. The blue trajectory represents a sample path in the CR3BP, constructed by selecting a nearby perijove within $\Pi_{L_{1,1}}^{t+}$, $\Pi_{L_{2,2}}^{t-}$; again, the red periapses from the ephemeris path very nearly cover the blue periapses. The ephemeris path of comet KM in Fig. 11b is described as an I-to-I itinerary, thus, the stable and unstable manifolds associated with the L_1 Lyapunov orbit are considered to examine the motion of the comet. A map depicting periapses along the stable manifold appears in Fig. 11a in addition to regions $\Pi_{L_{1,1-2}}^{t+}$; regions $\Pi_{L_{1,3-6}}^{t+}$ are also included but are nearly covered by crossings of the manifolds. The map associated with the unstable manifold and regions $\Pi_{L_{1,1-2}}^{t-}$ appears as the reflection of the map in Fig. 11a across the x -axis. Comet KM experiences 3.5 revolutions about Jupiter. From Fig. 11b, the last revolution appears to be about L_1 rather than Jupiter, however, all periapses within the region $x_{L_1} \leq x \leq x_{L_2}$ are considered and $p = 3.5$. Only the first and last passages of perijove are prograde. Selecting $m = 1$ and $n = 4$, periapsis regions $\Pi_{L_{1,1}}^{t+}$ and $\Pi_{L_{1,4}}^{t-}$ are depicted in the map in Fig. 11b. The planar projection of the indicated comet perijove lies close to the intersection $\Pi_{L_{1,1}}^{t+}$ and $\Pi_{L_{1,4}}^{t-}$. A representative periapsis within $\Pi_{L_{1,1}}^{t+}$ and $\Pi_{L_{1,4}}^{t-}$ is plotted in blue, and yields the sample CR3B path possessing the same qualitative behavior as comet KM. It should be noted that the trajectory for comet KM, as it appears in Fig. 11b from the HORIZONS data, differs from that generated by Ohtsuka et al. (2008). The HORIZONS propagation depicts KM approaching Jupiter from the interior region, rather than the exterior region. Periapses within the region $\Pi_{L_{2,4}}^{t-}$ lie just beside those within the region $\Pi_{L_{1,4}}^{t-}$, so

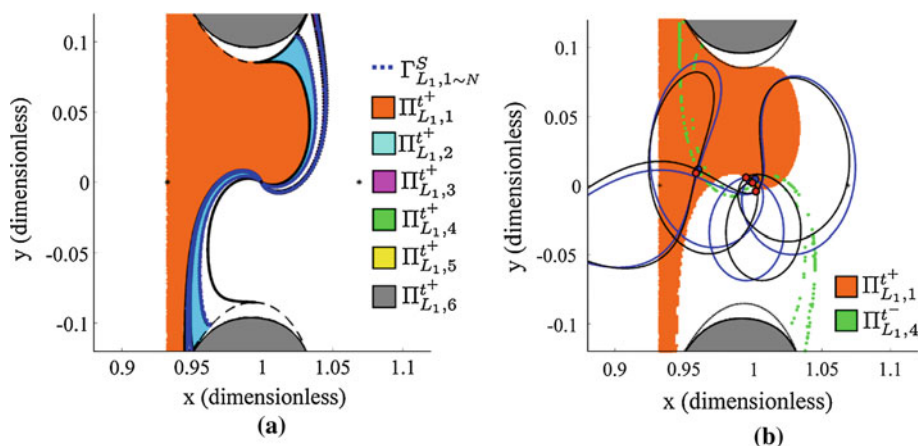


Fig. 11 Periapsis map associated with comet 147P/Kushida-Muramatsu ($C = 3.016$) with ephemeris path in black with red periapses; sample planar path approximation in blue with blue periapses, **a** $\Gamma_{L1,1\sim N}^S$, $C = 3.016$, **b** Path of KM

that a small perturbation in the final periapsis along the blue path in Fig. 11b yields an E-to-I transit.

By applying techniques previously demonstrated for trajectory design, the behavior of comets that experience TSC about Jupiter can be explored relative to the invariant manifolds. For each of the comets examined, the periapses along the ephemeris path closely follow the structures present in the periapsis maps for similar energy levels. It is straightforward to locate sample planar trajectories that closely resemble the ephemeris paths, demonstrating that the behaviors exhibited by these comets exist within the framework of the planar CR3BP.

6 Transit and temporary capture in the spatial CR3BP

While computing planar transit trajectories with predetermined behaviors is relatively straightforward, it is non-trivial to locate analogous solutions in the spatial CR3BP (SCR3BP). This challenge is addressed by Gómez et al. (2004), who employ the stable and unstable manifolds asymptotic to both periodic and quasi-periodic libration point orbits in the vicinity of L_1 and L_2 for a variety of trajectory design applications. The two-dimensional manifold contours, $\Gamma_{L_i,m}^{U/S}$, that appear on the Poincaré map for the PCR3BP are four-dimensional objects in the spatial problem. The authors reduce these objects to two dimensions by applying additional constraints, and Poincaré maps are employed to locate initial conditions that yield transit trajectories with specified behaviors. This strategy allows for the design of trajectories in the spatial problem using invariant manifolds. As an alternative, if the trajectories that lie within the manifolds, i.e., the transit trajectories, are located without computation of the manifolds themselves, then maps displaying regions $\Pi_{L_i,m}^{t\pm}$, such as those appearing in Fig. 9, can be employed to gain insight into the available solution space.

6.1 Identification of temporary capture trajectories possessing specified characteristics in the SCR3BP

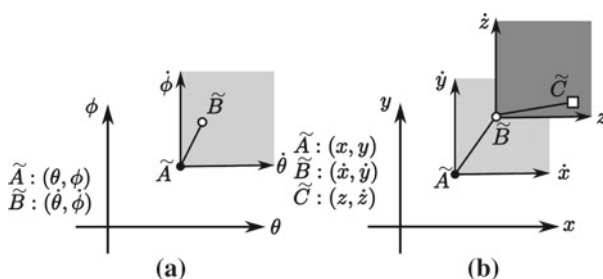
In this investigation, maps displaying regions $\Pi_{L_i,m}^{t\pm}$, are employed to gain insight into the available solution space and to locate trajectories with predetermined characteristics. Transit

trajectories in the spatial problem are located using a grid search, similar to that implemented in the planar case, and a strategy to display the periapses along these transits is explored. Rather than manifold periapses $\Gamma_{L_i,m}^{U/S}$, the regions $\Pi_{L_i,m}^{t^\pm}$ are employed to locate zones on the map corresponding to trajectories that exhibit distinct behaviors.

Consider the six-dimensional state associated with a passage of perijove in the spatial problem. The state is represented in spherical coordinates by five variables: the radius, ρ , from Jupiter, in-plane angle θ , out-of-plane angle ϕ , and angular rates $\dot{\theta}$, $\dot{\phi}$, noting that $\dot{\rho} = 0$ at periapsis. One variable is eliminated through a constraint on the value of Jacobi constant, and the state is fully defined through four variables. Thus, a periapsis map depicting $(\theta, \phi, \dot{\theta}, \dot{\phi})$ represents the full state associated with a periapsis, where ρ is computed from $C = 2U(\rho, \theta, \phi) - v^2$. Visualizing a four-dimensional map is challenging, however, many approaches exist to represent multivariate data points visually. Ward (2002) discusses a variety of representations that enable visual inspection of a multivariate data set. Pickett and Grinstein (1988) demonstrate the ‘stick-figure’ icon, formed by a series of connected line segments, to represent a multivariate data point. In this representation, an n -dimensional data point is represented by n connected segments; each of the n variables associated with the data point determine the orientation of one of the segments in the icon. This representation is easily adapted to represent states associated with crossings of four-dimensional Poincaré maps. For representations employed in this investigation, both length and orientation of the segments of the stick-figure are variable. Such map representations are applied to a variety of mission design scenarios by Haapala and Howell (2012). Consider the map in Fig. 12a, where the position data (θ, ϕ) is displayed as the solid point ($\tilde{A}$) in the $\theta - \phi$ coordinate plane. To represent the remaining states $(\dot{\theta}, \dot{\phi})$, a second coordinate frame is defined with point $\tilde{A}$ as the origin. The velocity state is defined within the $\dot{\theta} - \dot{\phi}$ coordinate plane, and is represented by the circle ($\tilde{B}$). Thus, the four-dimensional state is represented using the segment $\tilde{A} - \tilde{B}$ in Fig. 12a. By including one additional segment, each of the six Cartesian state variables is represented, as depicted in Fig. 12b. In this representation, the basepoint of the first segment is defined within the $x - y$ plane so that the planar projection of the periapsis position appears as the black point ($\tilde{A}$) at location (x, y) . Point $\tilde{A}$ serves as the base point for the second coordinate frame, defined as the $\dot{x} - \dot{y}$ plane, and the circle ($\tilde{B}$) represents the velocity states $(\dot{x}, \dot{y})$. A third coordinate plane is defined with $\tilde{B}$ as its origin, and the square marker ($\tilde{C}$) reflects the coordinates $(z, \dot{z})$. Then, the icon in Fig. 12b provides a visual representation for the full six-dimensional state associated with a map crossing. Maps utilizing these icons to display periapsis crossings prove useful for trajectory design in the spatial problem.

To compute periapsis maps associated with trajectories that experience TSC in the SCR3BP, grids of initial conditions in the libration point gateways are generated. Each grid is associated with a particular value of C , and corresponds to a particular x location, i.e., $x = x_{L_1}$ and $x = x_{L_2}$. For the particular value of x , the grid is generated in the $y - z$ plane

Fig. 12 Representations of higher-dimensional Poincaré maps, **a** representation of a four-dimensional map, **b** representation of a six-dimensional map



and, for each position (x, y, z) on the grid, the velocity magnitude, v , corresponding to the selected value of C is computed. The direction of the velocity is allowed to vary such that $\dot{x}$ is directed toward the P_2 region. Velocities are then sampled from the half-sphere of radius v , such that $\dot{x} > 0$ for $x = x_{L_1}$ and $\dot{x} < 0$ for $x = x_{L_2}$. Thus, each position on the y - z grid corresponds to a range of initial conditions, each with the same velocity magnitude but varying velocity directions. The initial conditions, around 150,000–200,000 total, are first integrated in reverse-time to verify that transition between the Jupiter region and the adjoining interior/exterior region occurs. Entry from the interior is defined by a crossing of the plane $x = x_{L_1} - 0.05$, and entry from the exterior is defined for trajectories that pass through the plane at $x = x_{L_2} + 0.05$. The trajectories that cross these planes are then propagated into the Jupiter region for a 500-year simulation. The resulting periapsis capture (reverse-time escape) trajectories are recorded. From the symmetry properties in the CR3BP, periapses along forward-time escapes are constructed by reflecting these periapsis states across the x - z plane such that $y \rightarrow -y$, $\dot{x} \rightarrow -\dot{x}$, and $\dot{z} \rightarrow -\dot{z}$.

Consider three-dimensional escape trajectories corresponding to the energy level $C = 3.0374$. The planar projection of periapses along three-dimensional L_1 escape trajectories appears in Fig. 13a, and along L_2 escapes in Fig. 13b. These maps serve as the spatial analog to the planar maps appearing in Fig. 4a, b. For clarity, L_2 periapses that occur for $x > 1.06$ (just within the L_2 gateway) are neglected in the counting process. No clear boundary exists to define L_1 gateway periapses, so all periapses are included when counting periapses prior to L_1 escape. Comparing the maps in Figs. 13 and 4, the structures present in the planar periapsis maps are very similar to those that appear in the planar projection of higher-dimensional periapsis maps. To demonstrate this similarity in structure, the planar projection of periapses within $\Pi_{L_1,1}^+$ and $\Pi_{L_2,1}^+$ from the spatial problem appear in Fig. 14a in addition to periapses along the planar Lyapunov manifolds; the planar extent of these first crossings of the higher-dimensional periapsis map is bordered by the planar manifold contours. An x - z projection of the periapsis regions $\Pi_{L_1,1}^+$ (right) and $\Pi_{L_2,1}^+$ (left) appears in Fig. 14b. The maximal out-of-plane component for the cluster of periapses $\Pi_{L_2,1}^+$ is $\sim 7.7 \times 10^{-4} = 6 \times 10^5$ km, indicating that E-to-I transits for $C = 3.0374$ are likely to possess a relatively small out-of-plane amplitude.

By increasing the energy level, the maximum z -amplitude for periapses within the region $\Pi_{L_2,1}^+$ increases and E-to-I trajectories possessing greater out-of-plane excursions are available. Consider transit trajectories corresponding to the energy level $C = 3.035$; the planar projection of the periapses along L_1 and L_2 escape trajectories for this energy level appears in Fig. 15. These plots serve as the spatial analog to those appearing in Figs. 8a, b, and similar structures are evident. The planar projection of periapses within $\Pi_{L_1,1}^+$ and $\Pi_{L_2,1}^+$ from the spatial problem appear in Fig. 16a, as well as periapses along the planar Lyapunov manifolds. Again, the planar manifold contours appear to bound the extent of the periapsis map near $z = 0$ for the first periapsis passages. This result may be expected, as the planar Lyapunov orbit bounds the extent of x - y plane crossings of the periodic and quasi-periodic orbits from the spatial problem (see Gómez et al. 2005). The x - z view of the regions $\Pi_{L_1,1}^+$ (right) and $\Pi_{L_2,1}^+$ (left) appears in Fig. 16b. The maximal out-of-plane component for periapses within the region $\Pi_{L_2,1}^+$ is increased to $\sim 6.4 \times 10^{-3} = 5 \times 10^6$ km, nearly an order of magnitude greater than for $C = 3.0374$. It is expected, then, that E-to-I trajectories possessing out-of-plane amplitudes on the order of 1×10^6 km are possible for $C = 3.035$.

To compute E-to-I trajectories in the SCR3BP, consider the six-dimensional Poincaré map representation that appears in Fig. 12b. The configuration $\hat{A} - \hat{B} - \hat{C}$ is plotted for each

Fig. 13 Projections of higher-dimensional Poincaré maps into planar configuration space for $C = 3.0374$; Jupiter $10\times$ actual size, **a** periapses along three-dimensional L_1 escapes, **b** periapses along three-dimensional L_2 escapes

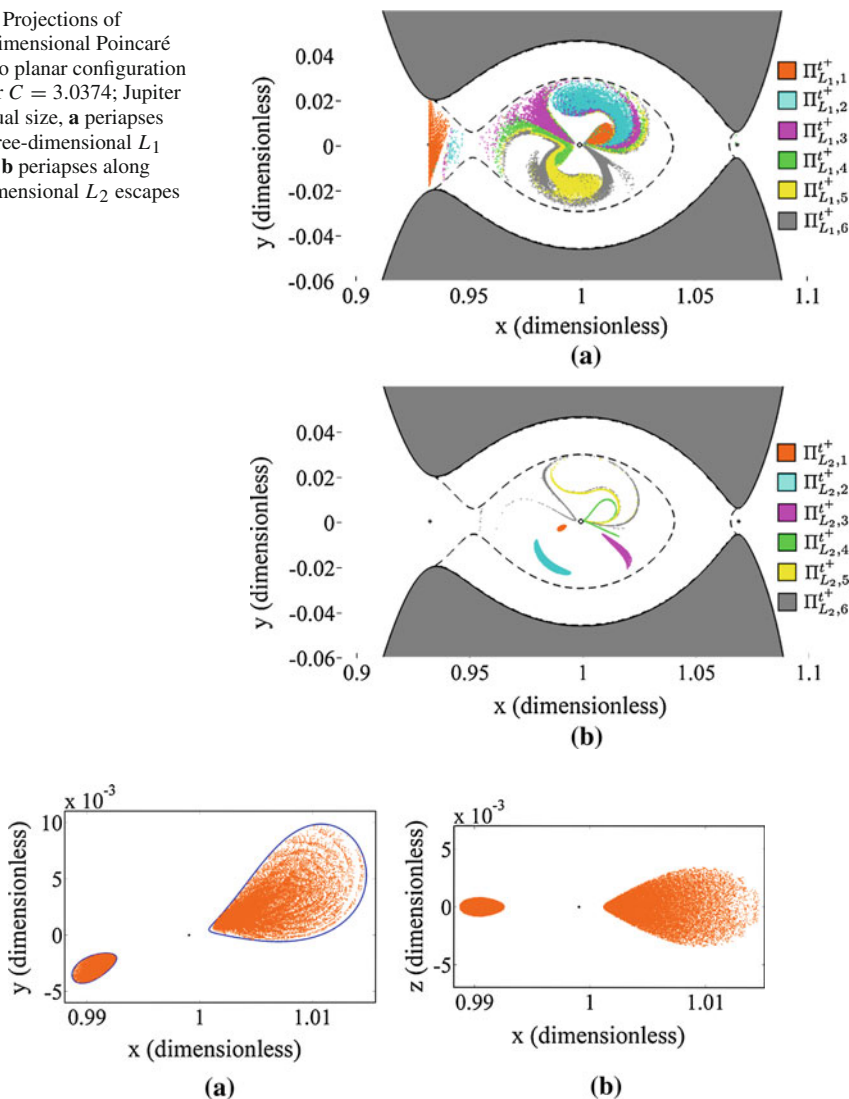


Fig. 14 Close view of regions $\Pi_{L1,1}^{t+}$ (right) and $\Pi_{L2,1}^{t+}$ (left) for $C = 3.0374$; Jupiter actual size, **a** close view of $\Pi_{L1,1}^{t+}$ (right) and $\Pi_{L2,1}^{t+}$ (left) with $\Gamma_{L1,1}^S$ and $\Gamma_{L2,1}^S$, **b** $x-z$ projection of $\Pi_{L1,1}^{t+}$ (right) and $\Pi_{L2,1}^{t+}$ (left)

periapsis, where $\tilde{A} : (x, y)$, $\tilde{B} : (x + k_1 \dot{x}, y + k_1 \dot{y})$, $\tilde{C} : (x + k_1 \dot{x} + k_2 \dot{z}, y + k_1 \dot{y} + k_2 \dot{z})$, where k_1 and k_2 are scaling factors. The map in Fig. 17 displays the regions $\Pi_{L1,1-6}^{t+}$ and $\Pi_{L2,1}^{t-}$ ($k_1 = 1 \times 10^{-3}$ and $k_2 = 5 \times 10^{-3}$). From this view, the orientation of each $\tilde{A} - \tilde{B} - \tilde{C}$ chain is not distinguishable, thus, it is useful to zoom-in on specific regions of interest. For example, Fig. 18a includes a zoomed view of the randomly selected black box (indicated with an arrow) from the region $\Pi_{L2,1}^{t-}$ in Fig. 17. In this view, the individual chains are distinguishable; for clarity, only perijoves for which $z > 0$ are displayed on the map. Assume that the goal is to generate E-to-I transit trajectories corresponding to $p = m + n - 3/2 = 1.5$ revolutions

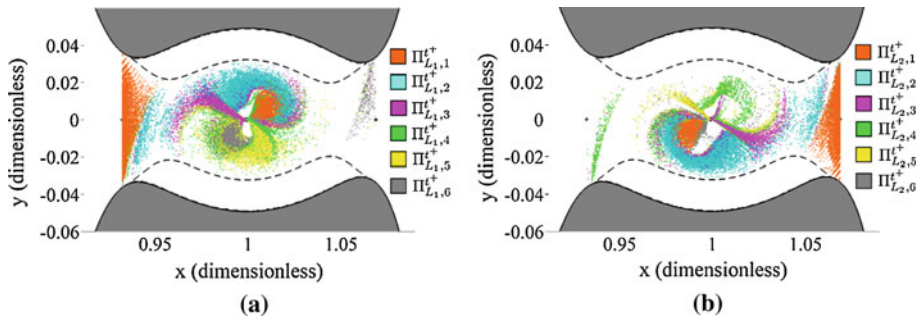


Fig. 15 Projections of higher-dimensional Poincaré maps into planar configuration space for $C = 3.035$; Jupiter $10\times$ actual size, **a** periapses along three-dimensional L_1 escapes, **b** periapses along three-dimensional L_2 escapes

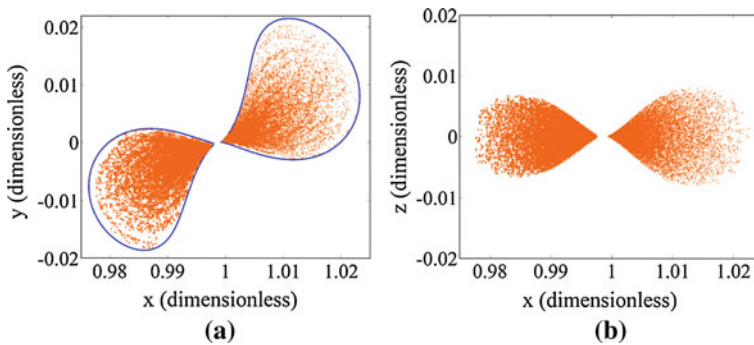


Fig. 16 Close view of regions $\Pi_{L_1,1}^{t+}$ (right) and $\Pi_{L_2,1}^{t+}$ (left) for $C = 3.035$; Jupiter actual size, **a** close view of $\Pi_{L_1,1}^{t+}$ (right) and $\Pi_{L_2,1}^{t+}$ (left) with $\Gamma_{L_1,1}^S$ and $\Gamma_{L_2,1}^S$, **b** x-z projection of $\Pi_{L_1,1}^{t+}$ (right) and $\Pi_{L_2,1}^{t+}$ (left)

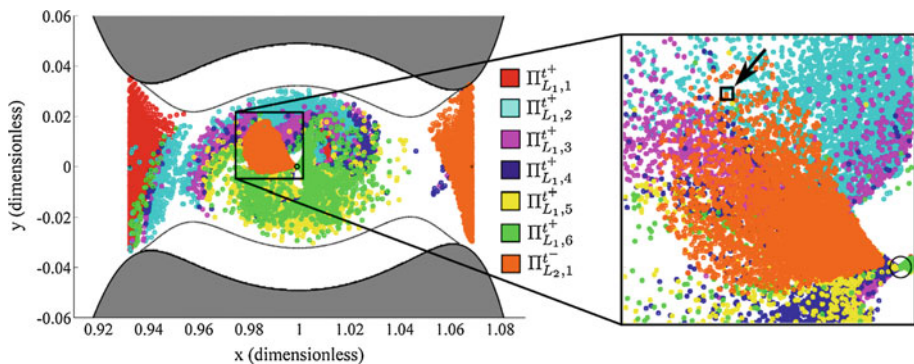


Fig. 17 Representation of a periapsis map displaying $\Pi_{L_1,1-6}^{t+}$ and $\Pi_{L_2,1}^{t-}$ for three-dimensional trajectories with $C = 3.035$; Jupiter $10\times$ actual size

about Jupiter. The maps are employed to locate members from the clusters $\Pi_{L_1,m}^{t+}$ and $\Pi_{L_2,n}^{t-}$ that possess similar states. Using the ‘stick-figure’ icons, the full six-dimensional states associated with periapsis map crossings are compared by visual inspection. For example, selecting $m = 2$ and $n = 1$, a ‘stick-figure’ that is a member of the cyan cluster $\Pi_{L_1,2}^{t+}$

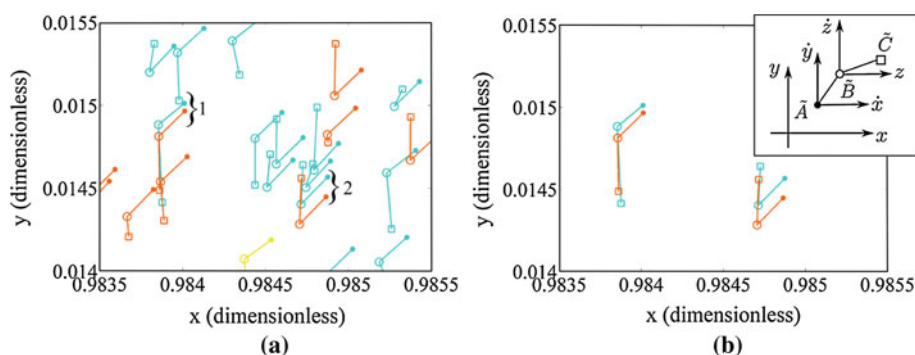


Fig. 18 Close view of a periapsis map representation to select $p = 1.5$ E-to-I transits for $C = 3.035$

is sought that is nearby and that possesses segments of similar length and orientation as a member of the orange $\Pi_{L_{2,1}}^-$ cluster. In Fig. 18a, two candidate pairs are located through visual inspection of the map and are labeled 1 and 2; these two pairs are isolated and plotted in Fig. 18b. In this isolated view, it is apparent that the segments for the pair of periapses labeled 1 differ more significantly than for pair 2, indicating that the states match more closely for pair 2. Without the application of a differential corrector, both periapses in pair 2 and the orange periapsis in pair 1 yield the specified characteristics, i.e., $p = 1.5$ E-to-I transit. The cyan periapsis from pair 1 leads to forward-time escape through the L_1 gateway after a second periapsis passage, as expected; propagating this periapsis in reverse-time, however, yields a trajectory that remains captured for approximately 38 years before escaping through the L_1 gateway. This arc possesses only one segment of the desired trajectory behavior, however, it is useful to seed an initial guess for a continuous E-to-I trajectory: the cyan member from $\Pi_{L_{1,2}}^+$ is propagated in forward-time until escape through the L_1 gateway occurs after passage through two periapses; the orange member of $\Pi_{L_{2,1}}^-$ is propagated in reverse-time yielding an immediate escape through L_2 . These two arcs are concatenated to form a discontinuous baseline, and continuity along the trajectory is achieved through a differential corrections process. A multiple-shooting approach is employed (see Betts 2005) wherein each arc is discretized and represented as a series of nodes or patch points. Small variations in the full state are applied at each node, in addition to variations in the propagation time between nodes, to satisfy continuity along the path to within a given tolerance ($1 \times 10^{-12} \sim 1$ m, 1×10^{-8} m s $^{-1}$). A continuous $p = 1.5$ E-to-I transit trajectory, corrected using the periapses from pair 1 in Fig. 18a, is plotted in Fig. 19a. An alternative method to locate transit trajectories with prescribed characteristics is demonstrated by Gómez et al. (2004), who compute transits by locating initial conditions that lie within the stable/unstable manifolds asymptotic to the center manifolds associated with the collinear points. To demonstrate the application of this method, the periodic halo and vertical orbits, as well as a subset of quasi-periodic tori, in the vicinity of L_1 and L_2 are computed as described in Sect. 3 for $C = 3.035$; to represent these libration point orbits, a map of the $z = 0$ crossings along each of the orbits computed is plotted in Fig. 19b. The stable/unstable manifolds associated with the periodic and quasi-periodic orbits are computed as described in Sect. 3. Note that only the northern halo and associated quasi-halos are computed; the manifolds associated with the southern orbits are obtained by a reflection across the x - y plane. A total of 29 Lissajous and 11 quasi-halo orbits about L_1 , and 49 Lissajous orbits in the vicinity of L_2 are computed (halo orbits about L_2 do not exist for this energy level). Approximately 1×10^6 manifold initial conditions are generated to approximate

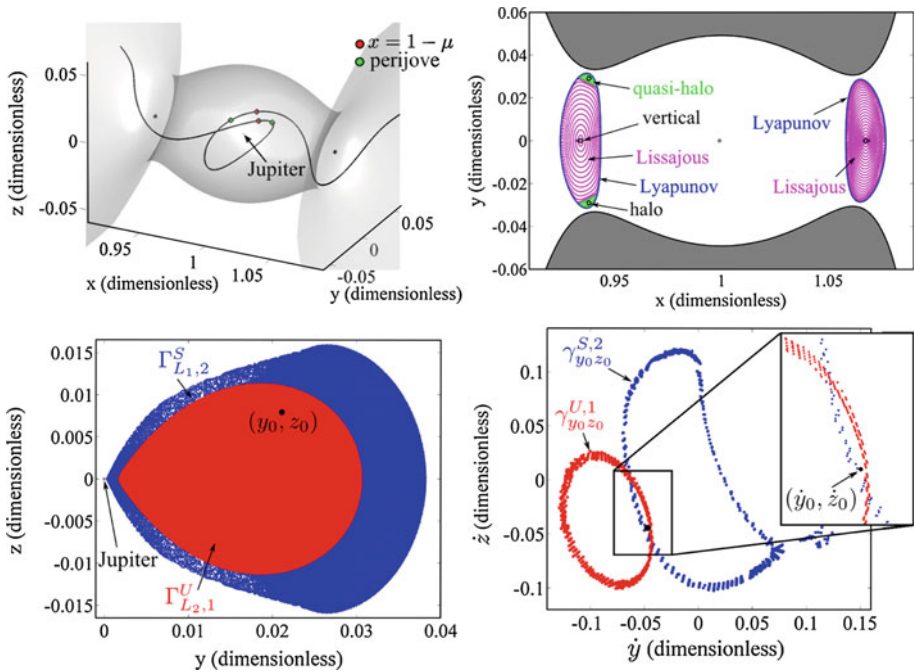


Fig. 19 Location of E-to-I transit relative to hyperbolic invariant manifold for $C = 3.035$

the hyperbolic invariant manifold asymptotic to the center subspace of each collinear point. To demonstrate that the trajectory in Fig. 19a exists within the appropriate stable/unstable manifolds, a Poincaré section is defined at $x = 1 - \mu$ for $\dot{x} < 0$ and the y - z projection of crossings $\Gamma_{L1,2}^S$ and $\Gamma_{L2,1}^U$ of the manifolds associated with L_1 and L_2 are examined in Fig. 19c. The first $x = 1 - \mu$ crossing after entry through L_2 along the path in Fig. 19a appears as a black point, defined as (y_0, z_0) , on the map within the region covered by $\Gamma_{L1,2}^S$ and $\Gamma_{L2,1}^U$. By selecting two additional parameters, i.e., $y = y_0$ and $z = z_0$, the manifold crossings are reduced to form contours. Because only a finite number of trajectories are employed to approximate the manifold, manifold crossings within a bounding box defined by $y_0 \pm \delta y$ and $z_0 \pm \delta z$ (here, $\delta y = \delta z = 3 \times 10^{-4} \approx 2 \times 10^5$ km) are considered to approximate the contour boundary formed by manifold crossings. These contours, labeled using the notation defined by Gómez et al. (2004) as $\gamma_{y_0 z_0}^{U,1}$ (unstable manifold) and $\gamma_{y_0 z_0}^{S,2}$ (stable manifold), appear in Fig. 19d. Any state defined on the map for $C = 3.035$ with $y = y_0$, $z = z_0$ and within $\gamma_{y_0 z_0}^{U,1} \cap \gamma_{y_0 z_0}^{S,2}$ represents an E-to-I transit of 1.5 revolutions about Jupiter. The point $\dot{y} = \dot{y}_0$ and $\dot{z} = \dot{z}_0$ associated with the black point in Fig. 19c is clearly within the intersection of the contours. In this example, a section defined by $x = 1 - \mu$ is employed because it is simpler to number crossings of the plane; many of the manifolds experience multiple passages of perijove before departing the vicinity of the libration point orbit from which the manifold is computed, making it difficult to consistently number crossings of the $\dot{r} = 0$ hyperplane.

6.2 Analysis of comet paths from the perspective of the SCR3BP

Shortly after the discovery of comet 11P/Helin–Roman–Crockett (HRC) in 1989, it was determined that the comet had experienced a close approach to Jupiter (Helin et al. 1989; Tancredi et al. 1990). Temporarily captured between 1967 and 1985, HRC experienced 6

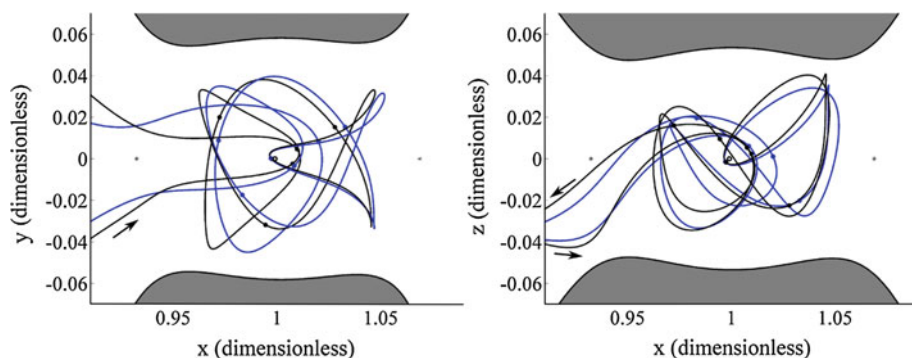


Fig. 20 Ephemeris path of comet 111P/Helin-Roman-Crockett (black) and a sample CR3BP path (blue) possessing similar characteristics as viewed in the Sun–Jupiter rotating frame; Jupiter $10\times$ actual size

passages of perijove and 5 of apojove within the 18.5-year TSC event, as apparent in Fig. 20. The jovicentric orbit is prograde for the duration of the capture. Hunter (1967) determined that jovicentric orbits with greater inclination tend to be more stable, measuring stability by the number of revolutions about Jupiter during TSC. The extended duration of the HRC capture may result from the high inclination of its orbit during the encounter.

To examine the transition of HRC between the interior and Jupiter regions, crossings of the surface of section defined by $x = 1 - \mu$ ($\dot{x} < 0$) are observed relative to the crossings of the stable and unstable invariant manifolds. A value of $C = 3.029$ is assumed for the energy level, and a sample set of periodic and quasi-periodic orbits within the center manifold associated with L_1 are computed. Approximately 1×10^6 initial conditions are computed for the unstable manifold and are integrated in the CR3B model. The stable manifold is determined by the transformation $y \rightarrow -y, \dot{x} \rightarrow -\dot{x}, \dot{z} \rightarrow -\dot{z}$. It is expected that the first crossing of HRC after entry from the interior lies within the first crossing of $W_{L_1}^{U+}$, and that the last crossing before escape from the Jupiter region is within the first crossing of $W_{L_1}^{S+}$. Define (y_i, z_i) as the (y, z) location at the i th, $i = 1, 2, 3, 4$, crossing of the path of HRC with the $x = 1 - \mu$ ($\dot{x} < 0$) hyperplane after entry from the interior. To locate the state along the comet path at the i th map crossing relative to the unstable manifold, the contour $\gamma_{y_i z_i}^{U,i}$ is computed for each crossing; to check for escape back to the interior, the contour $\gamma_{y_i z_i}^{S,1}$ is also located. The velocities $(\dot{y}_i, \dot{z}_i)$ along HRC are expected to lie within the corresponding contour $\gamma_{y_i z_i}^{U,i}$. If $(\dot{y}_i, \dot{z}_i)$ also lies within $\gamma_{y_i z_i}^{S,1}$, then HRC is expected to immediately escape and transition to the interior region. For $i = 1 \rightarrow 4$, the contours $\gamma_{y_i z_i}^{U,i}$ and $\gamma_{y_i z_i}^{S,1}$ appear in Fig. 21a–d. The second crossing along HRC appears to lie on the edge of the contour $\gamma_{y_2 z_2}^{U,2}$ in Fig. 21b, however, the reverse-time escapes, plotted in magenta, corresponding to the red contour actually lie along the boundary indicating that this manifold contour has a deformed “C” shape similar to those demonstrated by Koon et al. (2000). As the manifolds evolve and i increases, they tend to disperse causing $\gamma_{y_i z_i}^{U,i}$ become diffuse and to no longer appear as contours. The points $(\dot{y}_i, \dot{z}_i)$ along the path of HRC are plotted in black for comparison with $\gamma_{y_i z_i}^{U,i}$ and $\gamma_{y_i z_i}^{S,1}$. Clearly, $(\dot{y}_i, \dot{z}_i)$ lies outside of $\gamma_{y_i z_i}^{S,1}$ for $i = 1, 2, 3$, thus, the comet cannot escape to the interior region. For $i = 4$, $(\dot{y}_i, \dot{z}_i)$ is within $\gamma_{y_i z_i}^{S,1}$ resulting in immediate escape through the L_1 gateway. Because the manifold structures on the map become diffuse as the manifolds evolve, such as for $\gamma_{y_3 z_3}^{U,3} - \gamma_{y_4 z_4}^{U,4}$ in Figs. 21c, d, it is difficult to locate initial conditions ‘inside’ the manifold. For this reason, alternative methods to locate trajectories with specified behaviors are useful.

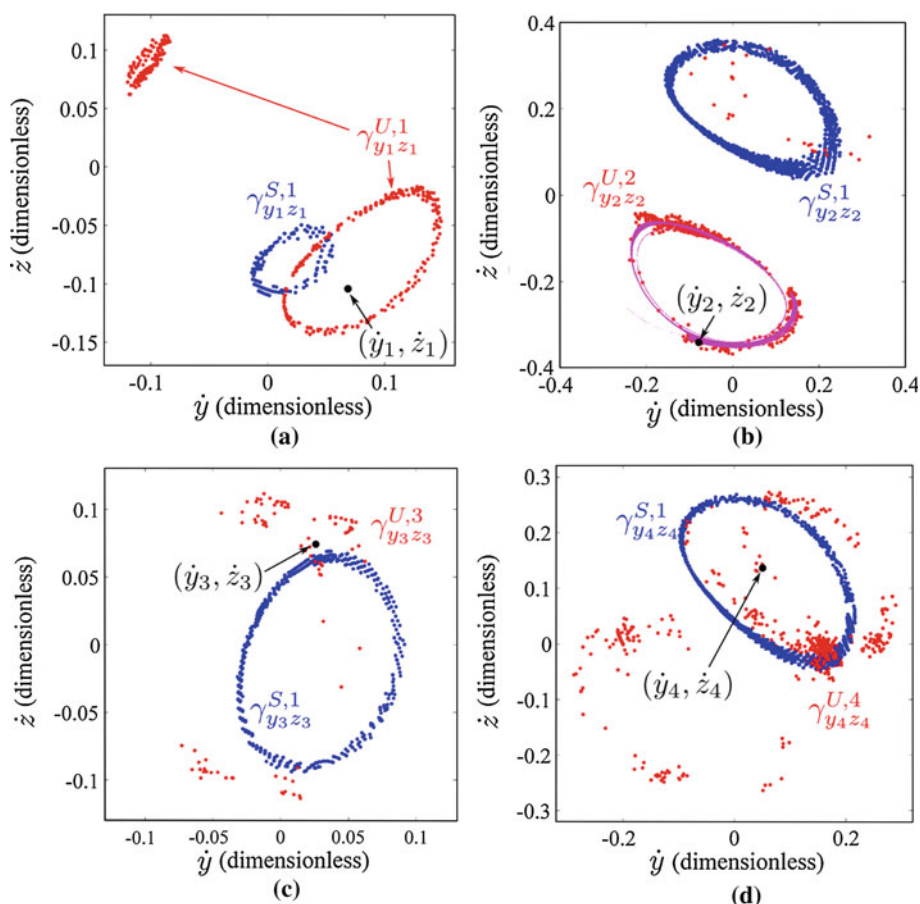


Fig. 21 Location of comet 111P/Helin-Roman-Crockett transit relative to invariant manifolds for $C = 3.029$, **a** HRC crossing 1, **b** HRC crossing 2, **c** HRC crossing 3, **d** HRC crossing 4

Periapsis maps displaying regions $\Pi_{L_1,m}^{t+}$, $\Pi_{L_1,n}^{t-}$ are helpful to locate an I-to-I transit matching the itinerary of HRC. To examine the location of the comet periapses relative to periapses on the map, the closest perijove to Jupiter is arbitrarily selected, i.e., $m = 4$ and $n = 3$, for display on the map. Regions $\Pi_{L_1,3}^{t-}$ and $\Pi_{L_1,4}^{t+}$ are viewed in the periapsis map appearing in Fig. 22a, where each periapsis is represented using the ‘stick-figure’ icon as depicted in Fig. 12b. A close view in the vicinity of the HRC periapsis appears in Fig. 22b, with the comet periapsis plotted and circled in black. One member each from the clusters $\Pi_{L_1,3}^{t-}$ and $\Pi_{L_1,4}^{t+}$ is selected such that the chains are located relatively close together and exhibit a similar orientation to one another as well as to that of the HRC periapsis. These selections are circled on the map in red and green, respectively, and yield trajectory segments that approximate the L_1 entry (propagating the red state in reverse-time) and L_1 escape (propagating the green state in forward-time) segments. These segments are concatenated, and continuity is enforced through a differential corrections process to yield the blue trajectory in Fig. 20. Utilizing the stick-figure icon to display crossings of Poincaré maps in the spatial problem provides an alternative method for trajectory design, especially useful in cases where long evolution times cause the map structures formed by crossings of the invariant manifolds

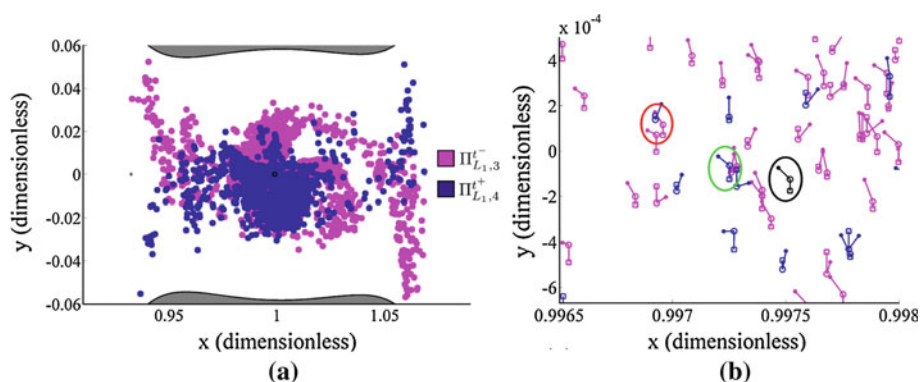


Fig. 22 Six-dimensional periapsis Poincaré map representation for $C = 3.029$, **a** six-dimensional map representation, **b** close view near comet perijove

to break down. Maps that display the full state associated with a map crossing are useful to reveal trajectories that closely resemble the ephemeris paths of comets in the SCR3BP. In this case, the behavior of comet HRC is clearly demonstrated within the framework of the CR3BP.

7 Conclusion

Temporary satellite capture of Jupiter family comets is a predictable phenomenon when considered from the perspective of the CR3BP. Transit between the interior/exterior regions of the ZVCs and the P_2 region has been widely investigated, and transit trajectories with a prescribed number of revolutions about the smaller primary have been demonstrated by employing Poincaré maps that display crossings of the invariant manifolds associated with libration point orbits in the CR3BP (Conley 1968; Koon et al. 2000; Gómez et al. 2004). Using these techniques, in combination with periapsis Poincaré maps, the behavior of Jupiter family comets P/1996 R2, 82P/Gehrels 3, and 147P/Kushida–Muramatsu during TSC is explored relative to the invariant manifolds associated with planar Lyapunov orbits. For each of the comets examined, the periapses along the ephemeris path closely follow the structures present in periapsis maps computed for similar energy levels to the comets. Sample planar trajectories that closely resemble the ephemeris paths are located, demonstrating that the motions exhibited by these comets exist within the framework of the planar CR3BP.

For comets with high inclinations relative to Jupiter, such as comet 111P/Helin–Roman–Crockett, it is of interest to examine the behavior relative to invariant manifolds in the spatial problem. The methodology defined by Gómez et al. (2004) is employed to compare Poincaré map crossings along the comet path with structures defined by the invariant manifolds associated with libration point orbits in the spatial model. The comet is demonstrated to lie within the higher-dimensional manifold ‘tubes’, thus explaining the observed motion for comet 111P/Helin–Roman–Crockett. An alternative approach to trajectory design in the spatial CR3BP, employing a visual representation for the six-dimensional states associated with periapsis map crossings, is introduced. This approach is demonstrated to generate a sample transit trajectory in the CR3BP that exhibits the behavior of comet 111P/Helin–Roman–Crockett during TSC.

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References

- Barden, B.: Application of dynamical systems theory in mission design and conceptual development for libration point missions. Ph.D. thesis, School of Aeronautics and Astronautics, Purdue University, West Lafayette, Indiana (2000)
- Betts, J.T.: Practical Methods for Optimal Control Using Nonlinear Programming. Society for Industrial and Applied Mathematics, Philadelphia, Pennsylvania (2005)
- Carusi, A., Valsecchi, G.B.: Numerical simulations of close encounters between Jupiter and minor bodies. In: Gehrels, T., Matthews, M.S. (eds.) *Asteroids*, pp. 391–416. The University of Arizona Press, Tucson (1979)
- Conley, C.C.: Low energy transit orbits in the restricted three-body problem. *Soc. Ind. Appl. Math. J. Appl. Math.* **16**, 732–746 (1968)
- Davis, D.C., Howell, K.C.: Trajectory evolution in the multi-body problem with applications in the Saturnian system. *Acta Astron.* **69**, 1038–1049 (2011)
- Davis, D.C., Howell, K.C.: Characterization of trajectories near the smaller primary in the restricted problem for applications. *J. Guid. Control Dyn.* **35**, 116–128 (2012)
- Everhart, E., Nirupama, R.: Changes in total energy for 392 long-period comets, 1800–1970. *Astron. J.* **75**, 258–272 (1970)
- Fernández, J.A.: *Comets: Nature, Dynamics, Origin, and their Cosmogonical Relevance*. Springer, Dordrecht, Netherlands (2005)
- Grebow, D.: Generating periodic orbits in the circular restricted three-body problem with applications to Lunar South Pole coverage. M.S. Thesis, School of Aeronautics and Astronautics, Purdue University, West Lafayette, Indiana (2006)
- Gómez, G., Jorba, A., Masdemont, J., Simó, C.: Study of the transfer from the Earth to a halo orbit around the equilibrium point L_1 . *Celest. Mech. Dyn. Astron.* **56**, 541–562 (1993)
- Gómez, G., Koon, W.S., Lo, M.W., Marsden, J.E.: Connecting orbits and invariant manifolds in the spatial restricted three-body problem. *Nonlinearity* **17**, 1571–1606 (2004)
- Gómez, G., Marcote, M., Mondelo, J.M.: The invariant manifold structure of the spatial Hill's problem. *Dyn. Syst. An Int. J.* **20**, 115–147 (2005)
- Haapala, A.F.: Trajectory Design using Periapse Maps and Invariant Manifolds. M.S. thesis, School of Aeronautics and Astronautics, Purdue University, West Lafayette, Indiana (2010)
- Haapala, A., Howell, K.: Trajectory design using periapse Poincaré maps and invariant manifolds. In: 21st AAS/AIAA Space Flight Mechanics Meeting, New Orleans, Louisiana (2011)
- Haapala, A., Howell, K.: Representations of higher-dimensional Poincaré maps with application to spacecraft trajectory design. IAF 63rd International Astronautical Congress, Naples, Italy (2012)
- Helin, E., Helin, R., Roman, B., et al.: Comet Helin–Roman–Crockett (1989b). *IAU Circ.* **4714**, 1 (1989)
- Heppenheimer, T.A.: On the presumed capture origin of Jupiter's outer satellites. *Icarus* **24**, 172–180 (1975)
- Heppenheimer, T.A., Porco, C.: New contributions to the problem of capture. *Icarus* **30**, 385–401 (1977)
- Horedt, G.P.: Capture of planetary satellites. *Astron. J.* **81**, 675–680 (1976)
- Howell, K.C., Davis, D.C., Haapala, A.F.: Application of periapse maps for the design of trajectories near the smaller primary in multi-body regimes. *J. Math. Probl. Eng., Special Topic: Mathematical Methods Applied to the Celestial Mechanics of Artificial Satellites 2012*, pp. 1–22 (2012). Article ID 351759
- Howell, K. C., Guzmán, J.J., Anderson, J.P.: Trajectory arcs with lunar encounters for transfers to small amplitude Lissajous orbits. In: 11th AAS/AIAA Space Flight Mechanics Meeting, Santa Barbara, California, (2001a)
- Howell, K.C., Marchand, B.G., Lo, M.W.: Temporary satellite capture of short-period Jupiter family comets from the perspective of dynamical systems. *J. Astronaut. Sci.* **49**, 539–557 (2001b)
- Hunter, R.B.: Motions of satellites and asteroids under the influence of Jupiter and the Sun-I. Stable and unstable orbits. *Mon. Not. R. Astron. Soc.* **136**, 245–265 (1967)
- Koon, W.S., Lo, M.W., Marsden, J.E., Ross, S.D.: Heteroclinic connections between periodic orbits and resonance transitions in celestial mechanics. *Chaos* **10**, 427–469 (2000)
- Kresak, L.: Dynamical interrelations among comets and asteroids. In: Gehrels, T., Matthews, M.S. (eds.) *Asteroids*, pp. 289–309. The University of Arizona Press, Tucson (1979)

- Marchand, B.: Temporary satellite capture of short period Jupiter family comets from the perspective of dynamical systems. M.S. thesis, School of Aeronautics and Astronautics, Purdue University, West Lafayette, Indiana (2000)
- Ohtsuka, K., Ito, T., Yoshikawa, M., Asher, D.J., Arakida, H.: Quasi-Hilda comet 147P/Kushida-Muramatsu. Another long temporary satellite capture by Jupiter. *Astron. Astrophys.* **489**, 1355–1362 (2008)
- Olikara, Z., and Scheeres, D.: Numerical method for computing quasi-periodic orbits and their stability in the restricted three-body problem. In: 1st IAA/AAS Conference on the Dynamics and Control of Space Systems, Porto, Portugal (2012)
- Paskowitz, M.E., Scheeres, D.J.: Robust capture and transfer trajectories for planetary satellite orbiters. *J. Guid. Control Dyn.* **29**, 342–353 (2006)
- Pickett, R., Grinstein, G.: Iconographic displays for visualizing multidimensional data. *Proc. IEEE Conf. Syst. Man Cybern.* **1**, 514–519 (1988)
- Tancredi, G., Lindgren, M., Rickman, H.: Temporary satellite capture and orbital evolution of Comet P/Helin–Roman–Crockett. *Astron. Astrophys.* **239**, 375–380 (1990)
- Villac, B.F., Scheeres, D.J.: Escaping trajectories in the Hill three-body problem and applications. *J. Guid. Control Dyn.* **26**, 224–232 (2003)
- Ward, M.O.: A taxonomy of glyph placement strategies for multidimensional data visualization. *Inf. Vis.* **1**, 194–210 (2002)
- Wiggins, S., Wiesenfeld, L., Jaffé, C., Uzer, T.: Impenetrable barriers in phase-space. *Phys. Rev. Lett.* **86**, 5478–5481 (2001)